What lessons can the Canadian judiciary learn from American courts regarding the impact of glyphosate use on individuals and the environment?

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Introduction

In 2015, the International Agency on Cancer Research (IARC) classified glyphosate, the most common ingredient in herbicides, as "probably carcinogenic to humans." [1] Canadian and United States regulatory agencies, however, have claimed that glyphosate does not pose any risk to the public or the environment when used as intended.[2] This scientific disagreement has resulted in civil litigation in which cancer patients have sought damages from Glyphosate Based Herbicide (GBH) manufacturers. Complainants have also pursued judicial review on the grounds that regulatory agencies erred in approving and re-approving Glyphosate and GBH's.

This paper argues that in the context of negligence actions against GBH manufacturers, American jurisprudence is a compelling persuasive authority for the Canadian courts, specifically in their analysis of causation. In the context of judicial review, the divergences in the enabling legislation between the U.S and Canadian regulators of GBH suggest that American law may provide limited insight in the Canadian context. However, the rulings in both jurisdictions have substantive similarities that may offer valuable guidance for future rulings. Specifically, in both jurisdictions, courts have held that the regulatory agencies failed to comply with their enabling statutes. In both jurisdictions, the courts also focused on the agencies' inability to justify their failure to adequately consider objections to GBHs, and demanded the agencies' provide further analysis.

The argument is broken down into four parts: Part I gives a brief overview of glyphosate, glyphosate-based herbicides, their regulation in Canada and the United States, and the scientific controversy surrounding the carcinogenic nature of glyphosate. The section further explores the

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[1] Guyton KZ, Loomis D, Grosse Y, El Ghissassi F, Benbrahim-Tallaa L, Guha N, et al. "Carcinogenicity of tetrachlorvinphos, parathion, malathion, diazinon, and glyphosate." (2015) 16(5):490–1. International Agency for Research on Cancer Monograph Working Group doi:10.1016/ S1470-2045(15)70134-8 PMID:25801782.

contrast between the IARC and regulatory agencies' findings on glyphosate's potential to cause cancer. Part II contextualizes the GBH civil lawsuits against Monsanto in the United States and considers if Canada can learn any lessons from the American litigation. Part III explores the judicial review of Canadian & American regulatory agencies involved in the approval of glyphosate products. Part IV explains the effects and potential limitations of litigation in this area.

I. The Glyphosate Question:

A. Background on Glyphosate and its Importance

Glyphosate was initially introduced as the active ingredient in Round-Up, a synthetic herbicide developed by Monsanto (now owned by Bayer) in the 1970s. After the patent on Round-Up expired, glyphosate became available in generic forms, known as GBHs.[3] Glyphosate works by inhibiting a specific enzyme pathway essential for plant growth, making it highly effective in controlling a wide variety of weeds.[4] Round-Up was the first herbicide that could kill both emerging weeds above ground and roots underground.[5] It was initially introduced as a broad-spectrum herbicide used on fields between growing seasons, as well as in ditches, railways, and other areas. It increased agricultural productivity and profits due to its efficacy. In the early 1980s, Monsanto developed 'Round-Up Ready' crops, including corn, soybeans, cotton, and canola, all of which were genetically modified to be resistant to glyphosate. This modification allowed farmers to spray Round-Up on crops during the growing season, enabling the crops to thrive without weed competition. This method for weed control resulted in increased

[3] Gabrielle Argimon-Cartaya, "SYMPOSIUM: AN UNEQUAL BURDEN: ENVIRONMENTAL JUSTICE: NOTE: 'Rounding up' Roundup: One Last Hope for Glyphosate Regulation" (2024) 78 Miami L. Rev. 602 at 610.
[4] *Ibid*.
[5] *Ibid*.

yields and decreased the need for additional inputs and tillage. The benefits of Round-Up Ready crops led to their rapid and widespread adoption, making Monsanto the largest seed corporation globally.[6] This trend continues to the present day; in the United States, for example, approximately 90% of corn, cotton, and soybean crops are glyphosate-tolerant,[7] and in Canada, 58% of pesticides used are GBHs.[8]

B. Regulation of GBH In Canada and the Untitled States

Canada and the United States both have regulatory agencies responsible for ensuring pesticides do not threaten public health. In Canada, the Pest Management Regulatory Agency (PMRA) regulates pesticides, empowered by the *Pest Control Products Act* (PCPA).[9] The PCPA requires the PMRA to evaluate pesticides before they are registered and to re-evaluate them within at least 15 years to "ensure they pose minimal risk to human health and the environment."[10] In the United States, the agency responsible for ensuring pesticides are safe for human health and the environment is the Environmental Protection Agency (EPA). The EPA is empowered to regulate and re-evaluate pesticides by the Federal Insecticide, Fungicide, and Rodenticide Act (FIRA).

The PMRA and EPA have aligned their policies, pesticide evaluation parameters, and methods.[11] Both bodies regulate GBH's, including Roundup, and have deemed them safe for

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[6] *Ibid* at 604.
[7] Jefferies, Danica, "A potentially cancer-causing chemical is sprayed on much of America's farmland. Here is where it is used the most.", *NBC News* (10 October 2022), online: https://www.nbcnews.com/data-graphics/toxic-herbicides-map-showing-high-use-state-rcna50052
[8] Marie-Hélène Bacon et al, "Poisoning Regulation, Research, Health, and the Environment: The Glyphosate-Based Herbicides Case in Canada." (2023) 11(2):121 Toxics. (https://doi.org/10.3390/toxics11020121) at 1.
[9] Health Canada, *Pest control product (pesticides) acts and regulations*, 2023. Online: https://www.canada.ca/en/health-canada/services/consumer-product-safety/pesticides-pest-management/public/protecting-your-health-environment/pest-control-products-acts-and-regulations-en.html
[10] *Ibid*.
[11] *Bacon*, *supra* note 8 at 3.

human health and the environment. However, the scientific community has, begun to question the conclusions of both regulators about the safety of glyphosate.

C. Glyphosate: Carcinogen or not?

There has been scientific disagreement over the potential carcinogenic properties of glyphosate since Roundup's invention. In 1985, the EPA conducted its first review of glyphosate and found benign kidney tumours in mice exposed to high levels of glyphosate, leading to the classification of glyphosate as "possibly carcinogenic to humans."[12] However, the following year, the EPA re-classified glyphosate as not carcinogenic, determining that there was no relationship between kidney tumours in mice and their glyphosate exposure.[13] Despite regulatory approval by government bodies worldwide, including the PMRA and EPA, there is a growing concern about the potential health effects of GBHs, particularly their carcinogenic effects. The controversy surrounding this issue gained more attention in 2015 when the IARC, a branch of the World Health Organization, classified glyphosate as "probably carcinogenic to humans" based on existing public research.[14]

The EPA and PMRA disagreed with the IARC's findings and continue to maintain that Glyphosate is unlikely to pose a cancer risk to humans. In 2016, in response to the IARC's findings the EPA declared that glyphosate was "not likely to be carcinogenic to humans" and there "are no risks of concern" to public health from glyphosate when used as labelled.[15] The PMRA published similar statements in 2017 in their re-evaluation of Glyphosate decision.[16]

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[12] *Argimon-Cartaya*, *supra* note 3 at 616.
[13] *Ibid*.
[14] *Guyton*, *Supra* note 1.
[15] EPA, *Glyphosate Issue Paper: Evaluation of Carcinogenic Potential*, 2016.
[16] Health Canada, *Glyphosate (*re-evaluation decision), Catalogue No H113-28/2017-1E (Ottawa: Health Canada, 28 April 2017).

Dr. Charles Benbrook, an American agriculture professor, identified three primary reasons for the opposing conclusions reached by the EPA and IARC, both well-regarded agencies. [17] First, although both agencies referenced some of the same studies, the IARC predominantly focused on peer-reviewed studies conducted by independent scholars. 70% of these independent studies yielded positive results. In contrast, the EPA largely relied on unpublished studies, many of which were industry-funded- a trend which Benbrook noted as a common practice of the EPA. [18] Second, the EPA's analysis was on technical glyphosate, whereas the IARC examined formulated GBHs.[19] Third and last, the EPA's assessment was based on the typical exposure of the general public and their dietary exposure, neglecting occupational exposure. In contrast, the IARC's evaluation encompassed data from both dietary and occupational exposure providing a more comprehensive view. [20]

The debate over the potential carcinogenic risk posed by glyphosate has become a predominant global issue, sparking varying opinions among regulatory agencies and the scientific community. The disagreement over glyphosate's cancer risk is not confined to a single institution or country; rather it highlights differing perspectives worldwide. In the United States for example, California designated glyphosate a known human carcinogen in 2017 under Proposition 65 and imposed warning labels on GBHs. However, a federal court ruled that imposing warning labels on glyphosate products infringed upon Monsanto's First Amendment rights. [21]

This controversy led to a wave of lawsuits across the U.S with claims alleging that Roundup has caused non-Hodgkin's lymphoma (NHL). As of March 2024, Bayer has paid approximately $11

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[17] Charles M.Benbrook "How did the US EPA and IARC reach diametrically opposed conclusions on the genotoxicity of glyphosate-based herbicides?" (2019) 31:2 ENV'T SCIS EUR (https://doi.org/10.1186/s12302-018-0184-7).
[18] *Ibid* at 6-7.
[19] *Ibid* at 3.
[20] *Ibid* at 2.
[21] See *Nat'l Ass'n of Wheat Growers v. Becerra*, 468 F. Supp. (3d) 1247 (E.D. Cal. 2020)

billion in verdicts and settlements, with 54,000 active lawsuits still pending. [22] In Canada, ligation is still in its infancy, with no cases reaching trial just yet. However, in 2023, a class action lawsuit was certified in Ontario, and more cases are in the works.[23]

Given that Canada is in the early stages of Roundup litigation, there is an opportunity to glean insights from the experiences of U.S. tort litigation, particularly regarding burden of causation plaintiffs face and the outcomes of tort litigation cases.

II. Tort Litigation

The IARC classification of glyphosate as a probable human carcinogen in 2015 sparked a series of civil actions against GBH manufacturers across the United States. Plaintiffs asserted strict liability, negligence, failure to warn, false advertising, and deceptive practices. In 2016 and 2017 three landmark legal cases were tried in California *Johnson v Monsanto Co,*[24] *Hardeman v Monsanto Co,*[25] and *Pilliod et al v Monsanto Co.*[26] These cases are particularly important because they were the first cases to be tried on this issue and the only cases as of March 2024 which have been heard by the appellant court. Therefore, they are the only cases which have written judicial decisions, which can be relied upon as compelling persuasive authorities in Canadian jurisprudence.

In each of these cases, the plaintiffs alleged their regular use of Round-Up caused their NHL, a type of cancer to which glyphosate exposure has been scientifically connected. The plaintiffs allege Monsanto acted negligently in fulfilling its duties of care to ensure plaintiffs did

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[22]Ronald Miller, "Monsanto Roundup Lawsuit Update" (6 April 2024) , online: https://www.lawsuit-information-center.com/rounduplawsuit.html#:~:text=Although%20these%20settlements%20account%20for,been%20filed%20in%20state%20court.

[23] *DeBlock v Monsanto Canada*, 2023 ONSC 6954 [*DeBlock*].

[24] (Cal App 1st Dist July 20, 2020) [*Johnson* 2020].

[25] (9th Cir Cal May 14, 2021) [*Hardeman*].

[26] (Cal App 1st Dist August 9, 2021)[*Pilliod*].

not suffer an unreasonable risk of NHL as a result of exposure to GBH. In all 3 cases, the jury and appellant courts found Monsanto liable. In *Johnson,* the jury found Monsanto acted with a "willful and conscious disregard of others' safety and with 'corporate malice' in continuing to market Roundup notwithstanding the possible link to NHL."[27] The appellant court held that the jury could have reached this conclusion based on the evidence presented at trial. The court specifically mentioned that since the 1980s, Monsanto had been aware of scientific studies suggesting glyphosate was genotoxic in a way that may lead to mutation and cancer. In response, a Monsanto toxicologist suggested further testing.[28] Furthermore, Monsanto's internal emails revealed that Monsanto was searching for someone who was comfortable with the genotoxic profile of glyphosate and who could be influential with regulators when genotoxic issues arise.[29] In sum, the appellant court allowed the jury's verdict that Monsanto's actions were indicative of corporate negligence. A similar result to *Johnson* occurred in *Hardman* and *Piliod*.

Since this trilogy of rulings, more than 100,000 people in the United States have claimed that exposure to Round-Up has caused their NHL. Bayer has resolved most of these claims, paying approximately $11 billion dollars in settlements.[30] As of March 2024, only 18 complaints have gone to trial and received a verdict, and many cases are still ongoing. Half of these cases resulted in a verdict for Monsanto/Bayer. In cases where the plaintiffs were successful, the damages have ranged from $80 million to over $2 billion. [31]

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[27] *Johnson 2020*, *supra* note 24 at 457.
[28] *Ibid* at 448-449.
[29] *Ibid at 437.*
[30] *Miller*, *supra* note 21.
[31] In *Hardeman* the jury awarded $80 million; in *Pilliod* the jury awarded $2.05 billion, both awards were lowed on appeal.

Monsanto appealed the *Hardeman* and *Pilliod* decisions to the United States Supreme Court, however, the Supreme Court denied certiorari in both cases.[32] As of March 2024, the Supreme Court of the United States has not granted certiorari in any Round-Up litigation cases.

A. Challenges with Tort Litigation: Causation

The specific causes of any cancer are impossible to identify with certainty. One of the biggest challenges faced by the plaintiffs suing Monsanto in negligence is establishing causality, i.e., demonstrating that GBHs are responsible for their cancer diagnoses. To succeed, the American courts have held plaintiffs must provide scientific evidence that illustrates GBHs have the potential to cause cancer (general causation) and then establish a direct causal link between their cancer and GBHs (specific causation). Each state has its own specific means of interpreting this general test.

In California, where a majority of the Round-Up lawsuits have been filed, plaintiffs need to show by a preponderance of evidence that the use of Round-Up was a "substantial factor" in causing their harm. [33] However, the California Courts have recognized that this burden is "especially troublesome" in cases alleging cancer given the complexities of tracing the orgins of particular cancerous growths. [34] The lack of scientific consensus makes it challenging for courts to definitively determine whether glyphosate is carcinogenic.

In Canada, the legal standard for causation differs slightly; plaintiffs must show that their cancer would not have developed "but for" their exposure to GBHs. [35] Nonetheless, insights into the broad concept of causation can be learned from American cases. Analyzing the criteria that

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[32] Rebekah Yeager-Malkin "US Supreme Court again refuses certiorari for Bayer, Monsanto" (28 June 2022), online: https://www.jurist.org/news/2022/06/us-supreme-court-once-again-refuses-certiorari-for-bayer-monsanto/
[33] *Johnson v Monsanto Co* (Cal Super Ct October 22, 2018) at 4 [*Johnson* 2018].
[34] *Ibid*.
[35] See *Clements v Clements*, 2012 SCC 32.

American courts have deemed sufficient for meeting the causation requirement could prove beneficial for future Canadian plaintiffs with similar claims. Additionally, the American analysis serves as a compelling, persuasive authority for Canadian jurisprudence, offering valuable precedents for establishing causation. In fact, the American causation analysis was cited by the Canadian judge's decision to certify the Ontario class action.[36]

The first American case to go to trial alleging that Round-Up caused the plaintiff's cancer was *Johnson.* The plaintiff, Mr. Johnson, had frequently used Round-Up in his job as a groundskeeper for a school. Mr. Johnson was certified to use Round-Up products and always used personal protective equipment. But, in 2014 an accident occurred which left him soaked in Round-Up through his protective gear. Six months after the exposure, he was diagnosed with NHL.[37]

At trial, the Jury was tasked with the complex scientific question of whether Mr. Johnson's exposure to glyphosate was a significant factor in causing his NHL. Ultimately, the jury ruled in favour of Mr. Johnson. Monsanto then filed a motioned for Judgment Notwithstanding the Verdict (JNOV), asking the court to find legal basis to overturn the jury's finding that the plaintiff's exposure to GBHs was a substantial factor in causing his NHL.[38] Although the case was appealed to the California Court of Appeal First Appellate District, that court did not address the issues of causation directly but rather focused on the legality of the damages awarded. Therefore, the Superior Court of California's remarks on causation in the JNOV is the only written opinion of causation in the *Johnson* case. Furthermore, causation is not

[36] *DeBlock*, *Supra* note 23.
[37] *Johnson 2020*, *supra* note 24 at 437.
[38] *Johnson 2018*, *supra* note 33.

addressed in many future decisions, as many of them are jury decisions, making the Superior Court of California's remarks very valuable precedent.

Mr. Johnson's general causation evidence focused primarily on the IARC classification of glyphosate as probably carcinogenic to humans. His specific causation evidence was largely based on the testimony of an oncologist.[39] Although the oncologist admitted that they couldn't identify the cause of NHL in most patients, they concluded that Mr. Johnson's exposure to GBH was likely responsible for his cancer through a causation analysis called differential diagnosis.[40] Differential diagnosis is a process in which a physician rules in all possible causes of a plaintiff's illness and then rules out the least plausible causes until the most likely cause remains.[41] In this case, the fact that Mr. Johnson was much younger than the average NHL patient raised a red flag that his cancer was not likely to be idiopathic and more likely to be caused by exposure.[42] The oncologist considered known risk factors and causes of NHL, such as age, race, immunosuppressant therapies, autoimmune diseases, skin conditions, occupation, occupational exposures, and viruses. Based on the differential diagnosis, the oncologist concluded that the plaintiff's only known risk factors were his race (African American) and exposure to GBHs. Therefore, the oncologist determined that GBHs were the most substantial contributing factor to the plaintiff's NHL.[43]

The court held that the jury was free to give weight to the oncologist's testimony that GBHs were a substantial factor in causing the cancer because there was no substantial evidence of an alternative explanation for the plaintiff's NHL.[44] The California Court of Appeal has

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[39] *Ibid* at 2.
[40] *Ibid* at 4-5.
[41] *Ibid* at 5.
[42] *Ibid*.
[43] *Ibid*.
[44] *Ibid* at 7.

accepted differential diagnosis as meeting the threshold test for admissibility, providing precedent for the Superior Court to reach this conclusion.[45] However, the Superior Court also noted that it does not resolve scientific controversies, and that is a matter for the jury to resolve.[46] This was a primary reason why the Court declined Monsanto's judgment notwithstanding the verdict concerning liability. As such, the court concluded a reasonable jury could have reached the conclusion Monsanto was liable in causing Mr. Johnson's NHL based on the evidence presented at trial.

In the subsequent cases *Hardman* and *Pilliod,* the plaintiffs also had expert witnesses who used differential diagnosis to show specific causation. In the *Pilliod* case on appeal, Monsanto argued the physicians who conducted differential diagnoses opinions "lacked reliable foundation" and were therefore "speculative and cannot constitute substantial evidence to support the verdicts".[47] The court disagreed and held that the experts did rule out alternative causes by either concluding they were not independent risk factors or explicitly testifying in their opinion the other factors were not a cause.[48] The success of the *Johnson, Hardman* and *Pilliod* trilogy suggests that relying on the IARC's classification to prove general causation and the reliance on differential diagnosis can be successful in showing specific causes.

Unfortunately, of the 9 cases that returned a verdict in favour of Monsanto, in 8 of the cases, the jury ruled this, and in the 9th, the judge granted a direct verdict without reasoning, so we are unaware of what reasons were insufficient for a causal link to be found.

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[45] *Ibid* at 6.
[46] *Ibid* at 7.
[47] *Pilliod*, *supra* note 26 at 624.
[48] *Ibid*.

B. Lessons Canada Can Learn

In its decision to certify *DeBlock* v *Monsanto,* the Ontario Superior Court examined the issue of causation, ultimately concluding that causation should not and cannot be determined at this stage, as there would need to be a more in-depth screening process. The judge highlighted the ongoing debate regarding the connection between glyphosate and cancer, referencing mixed outcomes in various U.S. court cases stating:

"There has been a great deal of litigation in the United States. Some individual actions have been tried there. Results have been mixed. In *Dewayne Johnson v. Monsanto Company*, the claimant was described as "a grounds manager for a school district and a heavy user of herbicides" who sued the manufacturer after contracting NHL. A jury awarded compensatory and punitive damages. While the jury award was reduced on appeal, Monsanto Company was ordered to pay more than $20 million U.S. in total on account of compensatory and punitive damages. Some, but not all, of the other claimants have been successful too". [49]

This acknowledgement of American litigation in the Canadian context suggests that such cases will likely influence forthcoming Canadian legal proceedings. Additionally, in the decision to certify the judge notes that the plaintiff relied on the IARC classification of glyphosate as probably carcinogenic and that Monsanto defended this classification stating that regulatory agencies around the world, including the PMRA, have re-evaluated glyphosate and repeatedly found it does not pose a threat to human health. It is worth noting that this is the exact same argument made by Monsanto in the three American cases above, but that the EPA re-evaluated glyphosate not the PMRA. Perhaps the Canadian court will look to its American counterparts for guidance on the role of the IARC classification in the causation analysis.

[49] *DeBlock*, *supra* note 23 at para 115.

III. Regulatory Agency Litigation

The controversy surrounding whether GBH causes cancer and its safety has also led to litigation against Canadian and American regulatory bodies. In these claims, the applicants argue that the re-evaluation of Glyphosate and approval of GBHs did not comply with the agencies' enabling Act. Unlike the tort claims, litigation on this issue has already taken place in Canada as well as in the United States. I will now explore the key cases in each jurisdiction.

A. EPA Litigation

The Environmental Protection Agency (EPA) is required to review each pesticide every 15 years under the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA). This is to ensure that the pesticide continues to satisfy the FIFRA standard for registration, that is, that the pesticide can perform its intended function without unreasonable adverse effects on human health or the environment. The FIFRA also establishes procedures and regulations the EPA must follow during the registration review.[50] The regulations require the EPA to assess any new information regarding the pesticide's potential risk to human health and the environment that has emerged since the EPA last issued the registration decision. Furthermore, the *Endangered Species Act* (ESA) requires federal agencies to ensure that the actions they authorize are not likely to jeopardize the continued existence of species listed as endangered under the ESA or destroy or adversely modify their designated habitat.[51]

In 2020, the EPA published their Glyphosate Interim Registration Review Decision. In this decision, the EPA maintained that glyphosate exposure does not cause any human health risks of concern and is "not likely to be carcinogenic to humans"; but did identify that glyphosate may

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[50] Federal Insecticide, Fungicide and Rodenticide Act, 7 U.S.C. §§ 136 (1947) s.3(g).
[51] Endangered Species Act, SO 2007, c 6, s.7(a)(2).

pose a risk to mammals, birds, and aquatic plants primarily from spray drift. However, the EPA's interim decision concluded that under FIFRA, the benefits of glyphosate outweigh the potential ecological risks. [52]

In *National Resources Defense Council v EPA*[53], two groups filed for review of the EPA Interim Decision. The first group, the Rural Coalition's petition, raised two issues with the Interim Decision. First, the EPA's conclusion that glyphosate did not pose a risk to human health was incorrect. Secondly, they insisted that the EPA should have followed the ESA's procedural requirements before issuing the Interim Decision. [54] The second group was the National Resources Defense Council (NRDC), which challenged the EPA's ecological risk assessment, cost-benefit analysis and risk mitigation requirements.[55]

The court first considered the human health challenge raised by the Rural Coalition. The court held that the EPA's conclusion in the Interim Decision was not consistent with the agencies own Guidelines for Carcinogenic Risks. The court noted that in the EPA's prior draft assessment on glyphosate (the "Cancer Paper") the EPA concluded that the association between glyphosate exposure and the risk of NHL could not be determined based on the available evidence. The court held that the EPA's conclusion that glyphosate is not likely to cause cancer based on the inability to reach a conclusion about the relationship between glyphosate and NHL does not pass the substantial evidence review standard. This is the administrative standard required under FIFRA. This standard requires the administrative record to show "such relevant evidence as a reasonable mind might accept as adequate to support the conclusion even if it is possible to draw

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[52] NRDC v. United States EPA, 38 F.4th 34, 2022 U.S. App. LEXIS 16821, 52 ELR 20070, 2022 WL 2184936 (9th Cir. June 17, 2022) [*NRDC*].
[53] *Ibid.*
[54] *Ibid* at 2.
[55] *Ibid.*

two inconsistent conclusions from the evidence." Furthermore, the court held the agency's reasoning must be internally consistent and coherent.[56] As a result, the court held that the EPA's classification of glyphosate as not likely to be a carcinogen did not meet the substantial evidence threshold thus the Court vacated the human health portion of the Interim Decision and remanded it for further analysis and explanation. [57]

The Court then addressed Rural Coalition's claim that the EPA impermissibly failed to follow the ESA. The court held that the EPA's registration review decision under FIFRA was an action that triggered the ESA's consultation requirement. It was undisputed that the EPA did not do so, thus the EPA violated the ESA. Nevertheless, the court declined to vacate that portion of the Interim Decision noting that the EPA must complete its final registration review which includes a formal consultation by October 2022 according to the timeline imposed by the FIFRA. Given that the final registration deadline was less than 6 months after the decision was released, shortening the EPA's time to consult would be very disruptive to the agency.

Lastly, the court considered the ecological risk assessment concerns raised by the NRDC. The court granted the EPA's motion to remand without vacatur as to the ecological portion of the interim decision but required the EPA to issue a new ecological portion by the October 2022 FIFRA deadline. The EPA could not meet the court-imposed deadline to review their ecological assessment; therefore, the EPA withdrew its Interim Decision and announced it is collaborating with other US environmental agencies to advance research on glyphosate's environmental impacts.

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[56] *Ibid*.
[57] *Ibid* at 4.

The EPA anticipates issuing a final registration review by 2026.[58] In the meantime, the regulations on glyphosate products are unaffected by this decision, and GBHs continue to be on the market.

In response to the *NRDC* decision and the EPA's withdrawal of the Interim Decision, Bayer stated, "We remain confident, based on the extensive science supporting its safety, that the agency will again conclude that glyphosate is safe for use and not carcinogenic as they have for decades." [59]

B. PMRA Judicial Review

In Canada, similar events have occurred; the regulatory agency released their re-evaluation, and an application was brought to the court to review the re-evaluation. In 2017 the PMRA re-evaluated glyphosate and granted continued registration of productions containing glyphosate for sale and use in Canada. [60] The PMRA stated in their re-evaluation decision an "evaluation of available scientific information found that products containing glyphosate do not present risks of concern to human health or the environment when used according to the revised label directions"[61]

After the PMRA released their re-evaluation decision, Safe Food Matters Inc. filed a notice of objection (NoO) in hopes that the Minister of Health would exercise their statutory discretion to appoint an independent review panel in accordance with the PCPA to consider Safe Food Matters‘ objections. In the NoO Safe Food Matters Inc. argued the PMRA failed to

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[58] EPA, Memorandum: Withdrawal of the Glyphosate Interim Registration Review Decision (21 September 2022) online https://www.regulations.gov/document/EPA-HQ-OPP-2009-0361-14447 .

[59] Todd Neeley, "Groups ask EPA to cancel Glyphosate registration", *Progressive Farmer* (2023). online: https://www.dtnpf.com/agriculture/web/ag/news/article/2023/12/13/environment-worker-groups-petition#:~:text=We%20remain%20confident%2C%20based%20on,of%20other%20expert%20regulators%20worldwide.%22

[60] Health Canada*, Re-Evaluation Decision: Glyphosate*, RVD2017-01 at 6. online: https://publications.gc.ca/collections/collection_2017/sc-hc/H113-28/H113-28-2017-1-eng.pdf

[61] *Ibid* at 7.

adequately consider the health risk posed by the use of glyphosate as a desiccant. Specifically, the NoO highlighted that when glyphosate is applied for desiccation purposes, the resulting residues in seeds often exceed the maximum residue limit (MRLs), raising significant health concerns, particularly in foods like beans and chickpeas. Additionally, the NoO raised concerns about the enforcement and product labelling issues. To support its objections Safe Food Matters Inc. referenced several scientific studies, government publications and Health Canada policy documents.

However, the PMRA dismissed the objections raised by Safe Food Matters' NoO and exercised its discretion not to establish a review panel.[62]

In 2019 Safe Food Matters Inc. applied to the Federal Court to review the PMRA's rejection of their NoO on the basis that the rejection was unreasonable. [63] Safe Food Matters submits that the PMRA decision was unreasonable for four reasons:

1. It failed to interpret the statutory scheme governing the criteria for assessing the NoO;
2. It did not comply with the statutory scheme, as property interpreted;
3. It failed to address the impact on individuals; and
4. It failed to address Safe Food Mater's evidence and submission.

The Federal Court dismissed the case, reasoning that the NoO did not show scientifically founded doubt about the validity of PMRA's evaluations. [64]

Safe Food Matters appealed to the Federal Court of Appeal, which allowed the appeal and held the PMRA's decision not to establish a review panel was unreasonable as they did not comply with the purview of their enabling statute, the *PCPA*.[65]

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[62] *Safe Food Matters Inc v Canada (Attorney General),* [2022] FCJ No 96, at para 9 [*Safe Food Matters Inc*]
[63] *Ibid*.
[64] *Ibid* at para 30.
[65] *Ibid* at para 57.

The Federal Court of Appeal confirmed that the standard of review is reasonableness. Thus, it was tasked with determining if the PMRA decision was reasonable, having regard to the reasonableness standard of review established by the Supreme Court of Canada in *Canada (Minister of Citizenship and Immigration) v Vavilov.*[66]

The Federal Court of Appeal noted that the PMRA does have the direction to establish a review panel, but it can only exercise its discretion once the 2 factors set out in the regulations are met. The two factors are: (a) whether the information in the notice of objection raises <u>scientifically founded doubt</u> as to the validity of the evaluations on which the decision was based and (b) whether the advice of expert scientists would assist in addressing the subject matter of the objection. [67]

The court held the PMRA does have the discretion to interpret the meaning of these factors, including the threshold of "scientifically founded doubt," but there must be evidence that these factors were considered. The Federal Court of Appeal held that these factors were not considered by the PMRA. Furthermore, the PMRA's discretion must comply with the rationale and purview of the Act. The Act's primary purpose is preventing "unacceptable risks to individuals and the environment from the use of pest control products." [68] The court concluded that the PMRA decision falls short of these fundamental requirements.

The Federal Court of Appeal quashed the PMRA decision and remitted the matter back to the PMRA for reconsideration in accordance with guidance provided by the Court. [69] The court provided that the PMRA needs to explain their decision, based on the interpretation of the

[66] 2019 SCC 65.
[67] *Ibid* at para 46.
[68] *Ibid* at para 51.
[69] *Ibid* at para 68.

legislation. [70] The court specifically suggested the PMRA should regard and communicate how it has regarded the text and its purpose of the preamble of the Act, the primary purpose of the Act, and definitions such as “health risk”, “acceptable risk”, “scientifically based approach”, “scientifically founded doubt”, when performing a review of notice of objection.[71]

The PMRA did reconsider Safe Food Matters’ notice of objection as mandated by the Federal Court of Appeal in September 2022 and once again rejected Safe Food Matters’ NoO.[72] Safe Food Matters holds the view that the PMRA did not “follow the Guidance of the Court, and made an unreasonable decision, went against the purpose of the Act, and created a new legal test that cannot be satisfied.”[73] As a result, Safe Food Matters filed an application for judicial review of the PMRA’s second decision to reject their NoO. The Federal Court has not yet released their decision on the preliminary hearing that occurred 2 years ago. As of October 2024, Safe Food Matters Inc. does not have an estimated time of arrival for the decision.

Safe Food Matters is involved in a second judicial review case. This case involved the PMRA’s renewal and registration for a GBH called Mad Dog Plus, (which is not manufactured by Bayer). [74]

In January 2023, Safe Food Matters applied to the Federal Court for judicial review of the decision of the PMRA to renew Mad Dog Plus. The applicants allege that the registration of Mad Dog Plus did not consider the recent scientific publications, specifically publications that address glyphosate’s potential harm to the environment and human health that the applicants sent the

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[70] *Ibid* at para 66.
[71] *Ibid* at para 65.
[72]Safe Food Matters, *“Issues: Glyphosate”* Online: https://safefoodmatters.org/issues-glyphosate/#:~:text=Health%20Canada's%20Pest%20Management%20Regulatory,again%20on%20September%2029%2C%202022.
[73] *Ibid* .
[74] *Friends of the Earth Canada v Canada (Attorney General*), [2023] FCJ No 1557.

PMRA during their review (known as the Ecojustice Letter). Safe Food Matters alleges the registration decision of the PMRA was unreasonable as they did not consider "complete and up-to-date information on potential environmental and health risks" including current published science reports on the risks of GBH. Safe Food Matters alleges that without considering this up-to-date science, the PMRA lacks reasonable certainty that no harm to the environment and human health will occur, thus making the renewal unreasonable. [75]

The PMRA states they are aware of the literature in the Ecojustice letter, however this does not change PMRA's assessment that the risks associated with glyphosate are acceptable but did not provide reasoning. The Federal Court held that it could not engage in a "meaningful judicial review" of PMRA's decision to renew Mad Dog Plus without understanding their rationale for concluding the Ecojustice Letter did not change their assessment.

The court then ordered the PMRA to send the applicants true copies of all written materials prepared for the renewal of Mad Dog Plus that address the scientific publications cited in the Ecojustice Letter. This case is still ongoing, and no further decisions have been made since this Court Order. [76]

C. Can we compare the regulatory agency review between Canada and the United States?

In both Canada and the United States, the regulatory agencies' re-evaluation of glyphosate has been challenged on the bias the agencies did not comply with their enabling legislation. In both jurisdictions, different standards of review are used. However, these standards of review are sufficiently substantively similar to justify similar outcomes.

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[75] *Ibid* at paras 1-7.
[76] *Ibid* at para 22.

In the United States, the court uses the substantial-evidence review standard, where the court considers whether there exists sufficient evidence to support the agency's factual determinations. In Canada, courts rely on the reasonableness standard, where the courts are not tasked with re-weighing the evidence but rather determine whether a regulatory agency's actions are transparent, justifiable and intelligible.

The fact that the standard of review and the legislation is different in the United States, when compared with Canada appears on its face to complicate the degree to which American judicial review can inform Canadian judicial review. But, perhaps the substantive similarities and outcomes of these judicial reviews, can provide a basis for informing rulings in the future. In both jurisdictions, under both standards of review, the courts acknowledge their role is not to evaluate the scientific evidence but rather to ensure that the agencies adhere to their enabling legislation when dismissing objections raised. In both countries, the courts were unsatisfied with the agency's analysis and stated the agencies needed to provide a better analysis. Perhaps the reason why the court was unsatisfied in both jurisdictions is that the agencies are relying on the same industry-funded scientific research in their re-evaluation.

IV. Conclusion: Is litigation driving reform?

A. Tort Litigation

The United States has experienced a significant amount of litigation related to glyphosate-based herbicides (GBHs). Many lawsuits have resulted in multi-million-dollar verdicts and billion-dollar settlements, yet GBH remains the most used herbicide in the United States. This raises the question of whether this litigation has impacted policy reform and if changes are needed. Cancer litigation is designed to settle claims between two parties and does not account for the potential widespread social harms caused by GBHs. Therefore, it is argued that it may have little impact on driving reform. However, in January 2023, Bayer removed its glyphosate-based herbicides from the US Lawn and Garden Market as a risk management strategy to prevent future litigations. This demonstrates that civil litigation has influenced Bayer, arguably not significantly, as Bayer's agriculture sector remains dominant. Furthermore, Bayer has stated on several occasions that they are committed to finding an alternative to glyphosate as an active ingredient in herbicides. Although Bayer claims this is due to the discovery of glyphosate-resistant weeds, it is likely also due, in part, to the controversy surrounding glyphosate's safety for human health and the environment. But this raises a question of if glyphosate's replacement will be any safer.

B. Agency Judicial Review

The judicial review of the regulatory agencies' decision to re-approve glyphosate is on track to being an endless merry-go-round of judicial review and subsequent reconsiderations. In Canada, the Federal Court hoped that their guidance would prevent this. However, subsequent action was brought less than a year later. In the United States, there were several cases against the EPA before the most recent case the *National Resources Defense Council v EPA* which was reviewed above. Definitive legal results and implications from any of these actions appear to be in the distant future, if ever. However, the outcome of the judicial review cases may have an impact outside the legal realm, as it is raising awareness of the potential harmful effects of glyphosate.